

1. 一均匀平面波从海水表面 ($x=0$) 沿 $+x$ 方向向海水中传播。在 $x=0$ 处, 电场强度为

$\vec{E} = \hat{y}100\cos(10^7\pi t)$ V/m, 若海水的 $\epsilon_r = 80$, $\mu_r = 1$, $\sigma = 4\text{S/m}$, 求:

(1) 衰减常数、相位常数、波阻抗、相速、波长、趋肤深度;

$$\tilde{\epsilon} = \epsilon \left(1 - j \frac{\sigma}{\omega\epsilon} \right) \approx -j \frac{\sigma}{\omega} = -j \frac{4}{10^7\pi} = -j1.27 \times 10^{-7} \quad \text{F/m}$$

$$\tilde{k} = \omega\sqrt{\mu\tilde{\epsilon}} = \beta - j\alpha = 2\sqrt{2}\pi - j2\sqrt{2}\pi \quad \text{rad/m}$$

$$\tilde{\eta} = \sqrt{\frac{\mu}{\tilde{\epsilon}}} = \pi e^{j\frac{\pi}{4}} \quad \Omega$$

$$v_p = \frac{2\rho}{b} = \frac{\sqrt{2}}{2} \quad \text{m/s}$$

$$d_p = \frac{1}{a} = \frac{\sqrt{2}}{4\rho} \quad \text{m}$$

(2) 写出海水中的电场强度和磁场强度表达式。

$$\vec{E}(x,t) = \hat{y}100e^{-2\sqrt{2}x} \cos(10^7\pi t - 2\sqrt{2}\pi x)$$

$$\vec{H}(x,t) = \hat{z} \frac{100}{\pi} e^{-2\sqrt{2}x} \cos(10^7\pi t - 2\sqrt{2}\pi x - \frac{\pi}{4})$$

2. 某种媒质中的相速为 $v_p = 2 \times 10^7 \sqrt[3]{\lambda}$ (m/s), 求 $\lambda = 10\text{m}$ 的电磁波的其群速度。

注: 由于此题没有明确指明波长是否为真空中的波长, 固本题给出了两种解法, 解法 1

认为波长为介质中的波长, 解法 2 认为波长为真空中的波长,

解法 1:

$$v_p = \frac{\omega}{k} = 2 \times 10^7 \sqrt[3]{\lambda} = 2 \times 10^7 \left(\frac{2\pi}{k} \right)^{1/3},$$

$$\omega = 2 \times 10^7 \sqrt[3]{\lambda} = 2 \times 10^7 \left(\frac{2\pi}{k} \right)^{1/3} k^2$$

$$\begin{aligned}
 v_g &= \frac{d\omega}{dk} = 2 \times 10^7 (2\pi)^{1/3} \frac{2}{3} k^{-1/3} \\
 &= 2 \times 10^7 (2\pi)^{1/3} \frac{2}{3} \left(\frac{2\pi}{\lambda} \right)^{-1/3} \\
 &= \frac{4}{3} \times 10^7 \sqrt[3]{\lambda}
 \end{aligned}$$

$$\text{帶入可得 } v_g = \frac{4}{3} \times 10^7 \sqrt[3]{10} = 2.87 \times 10^7 \text{ (m/s)}$$

解法 2:

$$\lambda = \frac{c}{f} = \frac{2\pi c}{\omega}, \text{ c 为真空中的光速}$$

$$\text{所以 } \frac{d\lambda}{d\omega} = -\frac{2\pi c}{\omega^2}$$

$$\text{而 } \frac{dv_p}{d\omega} = \frac{dv_p}{d\lambda} \frac{d\lambda}{d\omega} = \frac{2}{3} \times 10^7 \lambda^{-\frac{2}{3}} \times \left(-\frac{2\pi c}{\omega^2} \right) = -\frac{\lambda^{\frac{4}{3}}}{3\pi c} \times 10^7$$

$$\text{群速 } v_g = \frac{v_p}{1 - \frac{\omega}{v_p} \left(\frac{dv_p}{d\omega} \right)}$$

$$\text{帶入可得 } v_g = 0.75v_p = 1.5 \times 10^7 \sqrt[3]{10} = 3.23 \times 10^7 \text{ (m/s)}$$